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Roll No.



प्रश्न-पत्र कोड
Q.P. Code

55/S/3

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

भौतिक विज्ञान (सैद्धान्तिक) PHYSICS (Theory)

निर्धारित समय : 3 घण्टे

अधिकतम अंक : 70

Time allowed : 3 hours

Maximum Marks : 70

नोट	NOTE
(I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 27 हैं।	(I) Please check that this question paper contains 27 printed pages.
(II) कृपया जाँच कर लें कि इस प्रश्न-पत्र में 33 प्रश्न हैं।	(II) Please check that this question paper contains 33 questions.
(III) प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।	(III) Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
(IV) कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें।	(IV) Please write down the serial number of the question in the answer-book before attempting it.
(V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।	(V) 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

General Instructions :

Read the following instructions carefully and follow them :

- (i) *This question paper contains **33** questions. **All** questions are **compulsory**.*
- (ii) *This question paper is divided into **five** sections – **Sections A, B, C, D and E**.*
- (iii) *In **Section A** – Questions no. **1 to 16** are Multiple Choice type questions. Each question carries **1** mark.*
- (iv) *In **Section B** – Questions no. **17 to 21** are Very Short Answer type questions. Each question carries **2** marks.*
- (v) *In **Section C** – Questions no. **22 to 28** are Short Answer type questions. Each question carries **3** marks.*
- (vi) *In **Section D** – Questions no. **29 and 30** are case study-based questions. Each question carries **4** marks.*
- (vii) *In **Section E** – Questions no. **31 to 33** are Long Answer type questions. Each question carries **5** marks.*
- (viii) *There is no overall choice given in the question paper. However, an internal choice has been provided in few questions in all the Sections except Section A.*
- (ix) *Kindly note that there is a separate question paper for Visually Impaired candidates.*
- (x) *Use of calculators is **not** allowed.*

You may use the following values of physical constants wherever necessary :

$$c = 3 \times 10^8 \text{ m/s}$$

$$h = 6.63 \times 10^{-34} \text{ Js}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

$$\text{Mass of electron (} m_e \text{)} = 9.1 \times 10^{-31} \text{ kg}$$

$$\text{Mass of neutron} = 1.675 \times 10^{-27} \text{ kg}$$

$$\text{Mass of proton} = 1.673 \times 10^{-27} \text{ kg}$$

$$\text{Avogadro's number} = 6.023 \times 10^{23} \text{ per gram mole}$$

$$\text{Boltzmann constant} = 1.38 \times 10^{-23} \text{ JK}^{-1}$$

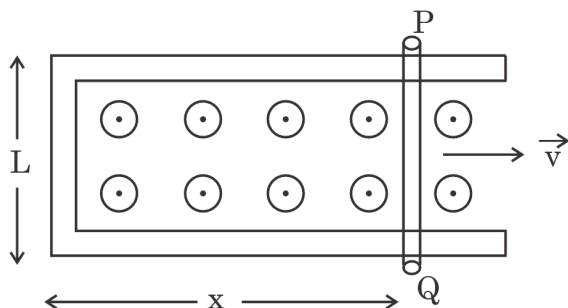
SECTION A

1. The electric field E associated with an electromagnetic wave is represented by

$$E_y = E_0 \sin(kx - \omega t)$$

Which of the following statements is correct ?

- (A) The wave is propagating along +x-axis.
 (B) The wave is propagating along +z-axis.
 (C) The magnetic field $\vec{B}$ of the wave is acting along +y-axis.
 (D) The magnetic field $\vec{B}$ of the wave is acting along -x-axis.
2. A capacitor of $5 \mu\text{F}$ is connected to an ac source of 200 V , $\frac{1000}{\pi} \text{ Hz}$ through a resistor of 100Ω . The phase difference between the voltage (V) applied and current (I) is :
- (A) 120° (B) 90° (C) 60° (D) 45°
3. m , m_n and m_p represents masses of ${}_Z^A\text{X}$ nucleus, a neutron and a proton, respectively. Then :
- (A) $m < (A - Z) m_n + Z m_p$ (B) $m = (A - Z) m_n + Z m_p$
 (C) $m = (A - Z) m_p + Z m_n$ (D) $m > (A - Z) m_n + Z m_p$
4. A rod PQ of resistance R lies across frictionless conducting rails in a constant uniform magnetic field $\vec{B}$, as shown in the figure. Assuming that the rails have negligible resistance, the force required to pull the rod to the right at constant velocity $\vec{v}$ is :

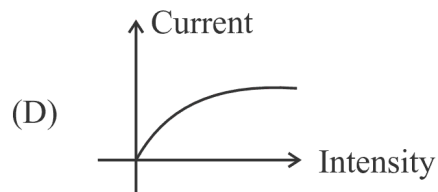
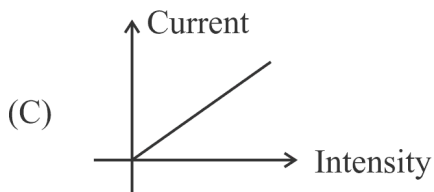
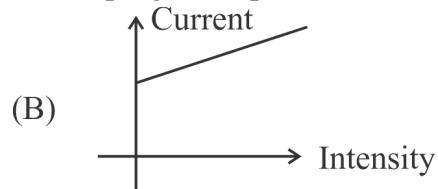
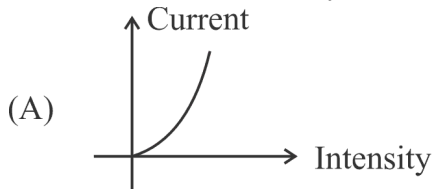


- (A) 0 (B) B/v
 (C) $\frac{B/v}{R}$ (D) $\frac{B^2 l^2 v}{R}$

5. When a ferromagnetic substance is heated to a temperature above its Curie temperature, it will :
- (A) behave like a diamagnetic material.
 - (B) behave like a paramagnetic material.
 - (C) permanently demagnetise.
 - (D) remain a ferromagnetic.
6. A point object is placed in air at a distance of $4R$ on the principal axis of a convex spherical surface of radius of curvature R separating two mediums, air and glass. As the object is moved towards the surface, the image formed is :
- (A) always real
 - (B) always virtual
 - (C) first virtual and then real
 - (D) first real and then virtual
7. A current of 5 A is flowing in a wire of length 1.5 cm . A force of 7.5 mN acts on it when it is placed in a uniform magnetic field of 0.2 T . The angle between the magnetic field and the direction of current is :
- (A) 30° (B) 45° (C) 60° (D) 90°
8. Two particles A and B of the same mass but having charges q and $4q$ respectively, are accelerated from rest through different potential differences V_A and V_B such that they attain same kinetic energies. The value of $\left(\frac{V_A}{V_B}\right)$ is :
- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 2 (D) 4

9. Which of the following statements is correct for alpha particle scattering experiment ?
- (A) For angle of scattering $\theta \approx 0$, the impact parameter is small.
 - (B) For angle of scattering $\theta \approx \pi$, the impact parameter is large.
 - (C) The number of alpha particles undergoing head-on collision is small.
 - (D) The experiment provides an estimate of the upper limit to the size of target atom.

10. Which one of the given graphs correctly represents the variation of photoelectric current with the intensity of incident radiation, keeping other parameters fixed ?



11. When intrinsic silicon semiconductor is doped with Al atom, then it :
- (A) decreases the number of holes in the conduction band.
 - (B) increases the number of holes in the valence band.
 - (C) increases the energy gap value.
 - (D) increases the number of electrons in the valence band.
12. Two cells of emf 2 V and 3 V, and internal resistance $1\ \Omega$ and $2\ \Omega$, respectively are connected in parallel. The effective emf and internal resistance of the combination cell are respectively :
- (A) 3 V, $1\ \Omega$
 - (B) 5 V, $3\ \Omega$
 - (C) $\frac{7}{3}$ V, $\frac{2}{3}\ \Omega$
 - (D) $\frac{6}{5}$ V, $\frac{2}{3}\ \Omega$

Questions number **13** to **16** are Assertion (A) and Reason (R) type questions. Two statements are given — one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (A), (B), (C) and (D) as given below.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).
- (C) Assertion (A) is true, but Reason (R) is false.
- (D) Both Assertion (A) and Reason (R) are false.

13. Assertion (A) : Although the surfaces of a goggle lens are curved, it does not have any power.

Reason (R) : In case of goggles, both the curved surfaces are curved on the same side and have equal radii of curvature.

14. Assertion (A) : The current density ($\vec{J}$) at a point in a conducting wire is in the direction of electric field ($\vec{E}$) at that point.

Reason (R) : A conducting wire obeys Ohm's law.

15. Assertion (A) : Nuclear fission reactions are responsible for energy generation in the Sun.

Reason (R) : Light nuclei fuse together in the nuclear fission reactions.

16. Assertion (A) : The torque acting on a current carrying coil is maximum when it is suspended in a radial magnetic field.

Reason (R) : The torque tends to rotate the coil on its own axis.

SECTION B

17. Explain the terms depletion layer and potential barrier for a junction diode and their formation. 2

18. An electron is passing through a region and experiences no force. Under what condition is it possible when the region has (a) only the magnetic field, and (b) both the electric and the magnetic fields ? Justify your answers. 2

19. (a) A cell is connected across an external resistance $12\ \Omega$ and supplies 0.25 A current. When the external resistance is increased by $4\ \Omega$, the current reduces to 0.2 A . Calculate (i) the emf, and (ii) the internal resistance, of the cell.

2

OR

- (b) Two point charges of $3\ \mu\text{C}$ and $4\ \mu\text{C}$ are kept in air at $(0.3\text{ m}, 0)$ and $(0, 0.3\text{ m})$ in x-y plane. Find the magnitude and direction of the net electric field produced at the origin $(0, 0)$.
20. A point light source rests on the bottom of a bucket filled with a liquid of refractive index $\mu = 1.25$ up to height of 10 cm . Calculate :
- (a) the critical angle for liquid-air interface
- (b) radius of circular light patch formed on the surface by light emerging from the source.
21. State Huygens principle. Using it draw a diagram showing the details of passage of a plane wave from a denser into a rarer medium.

2

2

2

SECTION C

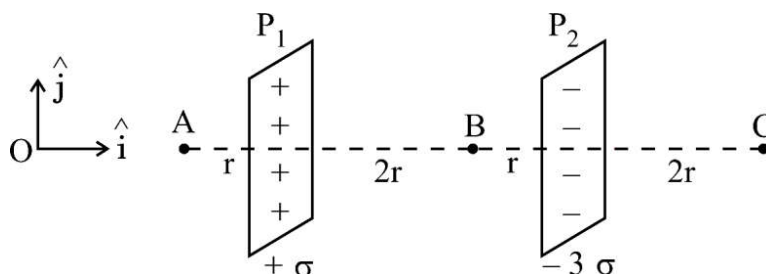
22. (a) The radius of a conducting wire AB uniformly decreases from its one end A to another end B. It is connected across a battery. How will (i) electric field, (ii) current density, and (iii) mobility of electrons change from end A to end B ? Justify your answer in each case.

3

OR

- (b) Two large plane sheets P_1 and P_2 having charge densities $+\sigma$ and -3σ respectively are arranged parallel to each other as shown in the figure. Find the net electric field ($\vec{E}$) at points A, B and C.

3



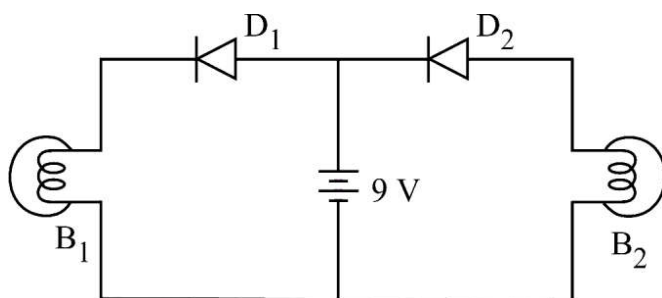
23. Two point charges of $10\ \mu\text{C}$ and $20\ \mu\text{C}$ are located at points $(-4\ \text{cm}, 0, 0)$ and $(5\ \text{cm}, 0, 0)$ respectively, in a region with electric field $E = \frac{A}{r^2}$, where $A = 2 \times 10^6\ \text{NC}^{-1}\ \text{m}^2$ and $\vec{r}$ is the position vector of the point under consideration. Calculate the electrostatic potential energy of the system. 3
24. Explain the following, giving proper reason : 3
- During charging of a capacitor, displacement current exists in the capacitor. But there is no displacement current when it gets fully charged.
 - The frequency of microwaves in ovens matches with the resonant frequency of water molecules.
 - Infrared waves are also known as heat waves.
25. Radiations of frequency $3.0 \times 10^{15}\ \text{Hz}$ are incident on a photosensitive surface of threshold frequency $1.0 \times 10^{15}\ \text{Hz}$. Calculate : 3
- work function of the surface
 - maximum kinetic energy of photoelectrons
26. A small circular loop of area $\frac{6}{\pi}\ \text{cm}^2$ is placed inside a long solenoid at its centre such that its axis makes an angle of 60° with the axis of the solenoid. The number of turns per cm is 10 in the solenoid. The current in the solenoid changes uniformly from 5 A to zero in 10 ms. Calculate the emf induced in the loop. 3
27. An ideal resistor R, an ideal inductor L and an ideal capacitor C are connected, one by one, across the same source of ac voltage $v = v_0 \sin \omega t$. It is observed that the same current I flows in each case.
- What will be the instantaneous value of current that will flow through their series combination when connected across the same source ?
 - How will the current in each case be affected if the frequency of the source is increased ?
- Justify your answers. 3
28. (a) Draw the shape of intensity distribution curve of the fringes due to diffraction at a single slit.
- (b) Derive the relation for the power of combination of two lenses placed in contact co-axially. 3

SECTION D

Questions number 29 and 30 are case study-based questions. Read the following paragraphs and answer the questions that follow.

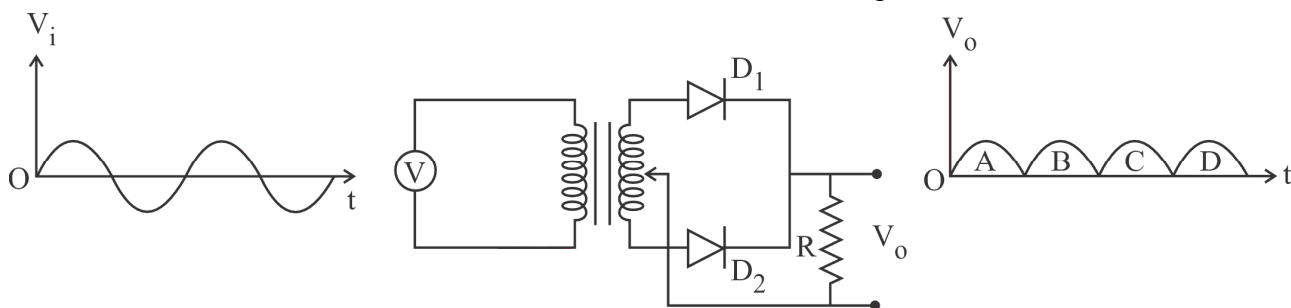
29. The process of converting ac into dc is called rectification and the device used is called a rectifier. When ac signal is fed to a junction diode during positive half cycle, the diode is forward biased and current flows through it. During the negative half cycle, the diode is reverse biased and it does not conduct. Thus the ac signal is rectified. The p-n junction diodes can be used as half-wave and full-wave rectifiers.

- (i) Which bulb/bulbs will glow in the given circuit ?



- (A) B₁ only
- (B) B₂ only
- (C) Both B₁ and B₂
- (D) Neither B₁ nor B₂

- (ii) (a) A full-wave rectifier circuit is shown in the figure. The contribution in output waveform from junction diode D_1 is :



- (A) A, D (B) A, C
(C) B, D (D) B, C

OR

(b) The output in a half-wave rectifier is : 1

- (A) unidirectional without ripple (B) steady and continuous
(C) unidirectional with ripple (D) steady but discontinuous

(iii) In a p-n junction diode, the majority charge carriers on p-side and on n-side are, respectively : 1

- (A) electrons, electrons (B) electrons, holes
(C) holes, holes (D) holes, electrons

(iv) If the frequency of the half-wave rectifier is 50 Hz, the frequency of full-wave rectifier is : 1

- (A) 25 Hz (B) 50 Hz
(C) 100 Hz (D) 200 Hz

30. Dipoles, whether electric or magnetic, are characterised by their dipole moments, which are vector quantities. Two equal and opposite charges separated by a small distance constitute an electric dipole, while a current carrying loop behaves as a magnetic dipole. Electric dipoles create electric fields around them. Electric dipoles experience a torque when placed in an external electric field.

(i) Two identical electric dipoles, each consisting of charges $-q$ and $+q$ separated by distance d , are arranged in x-y plane such that their negative charges lie at the origin O and positive charges lie at points $(d, 0)$ and $(0, d)$ respectively. The net dipole moment of the system is : 1

- (A) $-q d (\hat{i} + \hat{j})$ (B) $q d (\hat{i} + \hat{j})$
(C) $q d (\hat{i} - \hat{j})$ (D) $q d (\hat{j} - \hat{i})$

- (ii) E_1 and E_2 are magnitudes of electric field due to a dipole, consisting of charges $-q$ and $+q$ separated by distance $2a$, at points r ($\gg a$) (1) on its axis, and (2) on equatorial plane, respectively. Then $\left(\frac{E_1}{E_2}\right)$ is : 1
- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 2 (D) 4

- (iii) An electric dipole of dipole moment 5.0×10^{-8} Cm is placed in a region where an electric field of magnitude 1.0×10^3 N/C acts at a given instant. At that instant the electric field $\vec{E}$ is inclined at an angle of 30° to dipole moment $\vec{P}$. The magnitude of torque acting on the dipole, at that instant is : 1
- (A) 2.5×10^{-5} Nm (B) 5.0×10^{-5} Nm
(C) 1.0×10^{-4} Nm (D) 2.0×10^{-6} Nm

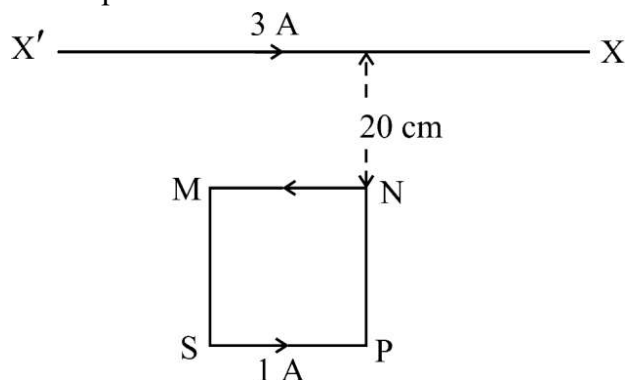
- (iv) (a) An electron is revolving with speed v around the proton in a hydrogen atom, in a circular orbit of radius r . The magnitude of magnetic dipole moment of the electron is : 1
- (A) $4 e v r$ (B) $2 e v r$
(C) $\frac{1}{2} e v r$ (D) $\frac{1}{4} e v r$

OR

- (b) A square loop of side 5.0 cm carries a current of 2.0 A. The magnitude of magnetic dipole moment associated with the loop is : 1
- (A) 1.0×10^{-3} Am² (B) 5.0×10^{-3} Am²
(C) 1.0×10^{-2} Am² (D) 5.0×10^{-2} Am²

SECTION E

31. (a) (i) With the help of a labelled diagram, explain the working of an ac generator. Obtain the expression for the emf induced at an instant 't'.
- (ii) A long, straight horizontal wire $X'X$ is held stationary and carries a current of 3.0 A. A square loop MNPS of side 10 cm, carrying a current of 1.0 A is kept near the wire $X'X$ as shown in the figure. Find the magnitude and direction of the net magnetic force acting on the loop due to the wire.



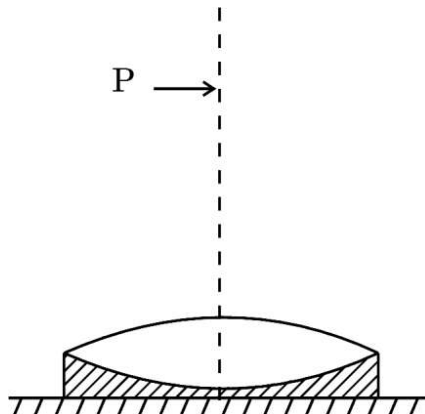
OR

- (b) (i) State Faraday's law of electromagnetic induction and mention the utility of Lenz's law. Obtain an expression for self-inductance of a coil in terms of its geometry and permeability of the medium.
- (ii) A resistance of $20\ \Omega$, a capacitance of $80\ \mu\text{F}$ and an inductor of $50\ \text{mH}$ are connected in series. This combination is connected across a $220\ \text{V}$ ac supply of variable frequency. When the frequency of supply equals the natural frequency of the circuit, calculate :
- (1) angular frequency of supply
 - (2) impedance of the circuit

32. (a) (i) What are the two main considerations for designing the objective and eyepiece lenses of an astronomical telescope ? Obtain the expression for magnifying power of the telescope when the final image is formed at infinity.
- (ii) A ray of light is incident at an angle of 45° at one face of an equilateral triangular prism and passes symmetrically through the prism. Calculate :
- (1) the angle of deviation produced by the prism
 - (2) the refractive index of the material of the prism
- 5

OR

- (b) (i) Describe a simple activity to observe diffraction pattern due to a single slit.
- (ii) The figure below shows an equiconvex lens (of refractive index 1.50) in contact with a liquid layer on top of a plane mirror. A small needle with its tip on the principal axis is moved along the axis until its inverted image is found at the position of the needle. The distance of the needle from the lens is measured to be 45.0 cm. When the liquid is removed and the experiment is repeated, the new distance is 30.0 cm. Find the refractive index of the liquid.
- 5



33. (a) (i) What are matter waves ? A particle of mass m and charge q is accelerated from rest through a potential difference V . Obtain an expression for de Broglie wavelength associated with the particle.
- (ii) Monochromatic light of frequency 5.0×10^{14} Hz is produced by a source of power output 3.315 mW. Calculate :
- (1) energy of the photon in the beam
 - (2) number of photons emitted per second by the source

5

OR

- (b) (i) State Bohr's postulates and derive an expression for the energy of electron in n^{th} orbit in Bohr's model of hydrogen atom.
- (ii) Calculate binding energy per nucleon (in MeV) of $^{12}_6\text{C}$.

5

Given :

$$m \left(^{12}_6\text{C} \right) = 12.000000 \text{ u}$$

$$m \left(^1_0\text{n} \right) = 1.008665 \text{ u}$$

$$m \left(^1_1\text{H} \right) = 1.007825 \text{ u}$$